

Ethan Huber

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Skills

Hardware & Design: KiCad, Altium, Soldering, Oscilloscopes, Multimeters, Power Supplies, OnShape, SolidWorks

Embedded & Protocols: STM32, ESP32, FPGA, Raspberry Pi, CAN Bus, VHDL

Software: Go, C, Java, C#, TypeScript, JavaScript, Python, Git, Linux, Docker

Education

Bachelor of Engineering, Computer Engineering (Co-op) *Sept 2024 – May 2029 (Expected)*

University of Guelph; 87% average (GPA: 3.7) *Guelph, Ontario*

- Relevant courses: Digital System Design, Computer Organization, Engineering Systems Analysis, Data Structures and Algorithms, Object Oriented Programming for Engineers, Software Systems Development and Integration

Experience

Systems Engineering Co-op Analyst *May 2026 – Aug 2026*

Toyota Motor Manufacturing Canada *Cambridge, Ontario*

- Reverse engineered a proprietary torque controller protocol and built a custom adapter to capture live sensor data from the hardware, working with vendors, technicians, and engineers to define requirements
- Co-led a **Python** ML anomaly detection system for real-time defect detection on live torque controller data
- Built a TypeScript/Playwright test automation framework covering 254 features

Process Control Team Member

May 2025 – May 2026

Toyota Motor Manufacturing Canada *Cambridge, Ontario*

- Supported final assembly using Toyota Production System principles and identified 19 defects during vehicle shipping preparation, preventing quality issues before customer delivery

Electrical Team Lead

Sep 2025 – Present

University of Guelph Robotics Team *Promoted to Lead in May 2026*

- Lead the electrical sub-team, coordinating with the mechanical and software teams to develop, test, and integrate the rover's electrical systems, from power distribution to CAN bus, for the CIRC
- Designed a 7-channel power distribution board (24V input to 19 terminals across 12V/5V/24V) with ideal diode controllers, and designed a stacked converter tray to meet thermal and footprint constraints
- Designed an **STM32**-based fan controller board with **CAN bus**, 2 NTC temperature probe inputs, and 2 daisy-chainable fan banks; ordered from JLCPCB and hand-assembled by hotplate
- Debugged CAN communication and verified rail stability under varying loads using oscilloscopes, multimeters, signal generators, and bench supplies; contributed to safety documentation for competition compliance

Frame Team Member

Sept 2024 – May 2026

Gryphon Racing *Guelph, Ontario*

- Designed motor controller mounts in **SolidWorks** and performed finite element analysis to optimize strength
- Created 8 **3D-printed** jigs to improve fabrication accuracy and assembly efficiency

Projects

Modular Agricultural Robot Platform // April 2026 – Present

[Link to project on portfolio](#) 🔗

- Designed and built a custom **STM32** control board with 4 MOSFET motor drivers for real-time motor control, supporting **UART** and **I2C** for modular tool and sensor expansion
- Engineered a repairable platform with a sheet metal aluminum chassis, custom **3D printed** in-hub gearboxes, modular sensor bumpers, and a swappable payload mount
- Developing GPS + IMU guided autonomous navigation with computer vision on the **Raspberry Pi** to scout corn fields and return data autonomously
- Built an Angular web-based mission manager for remote monitoring and control of robots

PCB Business Card // Dec 2025 – Jan 2026

github.com/zushiEdu/BusinessCard 🔗

- Designed a PCB in **KiCad** around an **STM32** driving a 12x12 LED matrix with 6 push-button inputs, and hand-soldered all SMD components

Filedex - file search engine // Jul 2025 – Sep 2025

github.com/zushiEdu/filedex 🔗

- Developed a semantic file search engine in **Go** for file indexing and retrieval, using **SQLite** for metadata storage
- Implemented graph traversal algorithms to identify relationships between tagged files
- Deployed local AI/ML recognition using Ollama to automate metadata tagging generation